INTERACTIONS BETWEEN RISKS

Learning Outcome

- identify financial and non-financial sources of risk and describe how they may interact

In some cases, a risk classified into one category could easily have been classified into another. Indeed, the interactions between risks are numerous. It has been said that market risk begets credit risk, which begets operational risk. That is, given unexpected market moves, one party then owes the other party money. Given the debtor–creditor nature of the relationship, the two parties must have internal operations that process the transactions and pay or collect the money. Thus, whenever there is credit risk, there is settlement risk. If there were no market risk, the other risks in the chain would likely be relatively minor. Legal risk often arises from market or credit risk. Large market moves create losses for one party. There is a long history of parties searching for loopholes in contracts and suing to avoid incurring the loss.

One simple example of an adverse risk interaction is counterparty risk. When trading a derivative contract, it is important to consider the cost of counterparty risk. Suppose Party A buys an out-of-the-money put option with a strike price of ¥1000—a contract theoretically worth ¥100 entitling him to as much as ¥1000 from Counterparty C if an underlying equity index is down. But there is a 2% chance that C could default; and assume that the possibility of default is considered independent of the performance of the equity market. This transaction, with payoffs adjusted for the possibility of default, might price at, say, ¥98 to A. But in reality, the credit risk of C's default is likely dependent on the equity market return. If the probability distribution of default risk overlaps substantially with that of the market being down, which is a likely scenario, then the risks interact, and the cost of risk is higher. In this example, perhaps the probability of C defaulting is 10% or more when the put option is in the money. So, A's expected payoff is lower as a result of facing a credit risk that is compounded by market risk. In fact, it is quite likely that in the extreme event—a deep decline in the equity market when A would presumably receive ¥1000—Party A will in all likelihood get nothing. Thus, the investor bears much more risk than initially thought as a result of the failure to consider the interaction of the two risks. And in doing so, Party A overpaid for the contract. This sort of risk interaction is so common in markets that practitioners have given it a very fitting term—"wrong-way risk." In fact, it was extremely common in the financial crisis of 2008, when holders of many securities based on mortgage credit believed that the risks were well-diversified when in truth, the risks were quite systematic.

Another example of interacting risks was experienced by many banks, funds, and private investment partnerships in 2008, as well as the hedge fund Long-Term Capital Management in 1998. Leverage, which manifested itself in higher market risk, interacted in an extremely toxic manner with liquidity risk and solvency risk and impaired or shuttered many investment firms.[18]

In most adverse financial risk interactions, the whole is much worse than the sum of its parts; the combined risk compounds the individual risks in a non-linear manner. For this example, a 2× levered organization might produce a 2% loss when its unlevered twin or baseline risk bears a 1% loss. If liquidity is a serious issue for the organization, then at a 10% baseline loss, the organization might face some moderate distress from liquidity or funding problems that it ends up losing 25% instead of 20%. It would not be surprising if this organization failed at a 30% baseline loss because of the toxic interplay between levered risk and liquidity problems. This resulting non-linear reaction to risk drivers exists

Earlier, we briefly described a common example of interacting risks for individuals. Suppose an individual works for a publicly traded company and, through an incentive program, receives shares of the company in her company retirement portfolio or for her personal holdings. Company policies may require that employees hold on to these shares for a number of years. When that time has elapsed, however, many individuals fail to recognize the incredibly concentrated risk they are assuming, so they hold on to their shares. An employee's reasoning for not selling the shares is often that the company she works for has been a solid performing company for many years, so she feels no reason to worry. Moreover, the team spirit often imbued in employees generates pride that can make employees believe that there is no better place in which to work and to invest their money. But if something goes wrong in the company or the industry, the employee may lose her job *and* her savings—an incredibly adverse interaction between market risk and human capital risk. The 2003 collapse of Enron remains a powerful historical example, with many loyal and honest employees losing virtually all of their retirement savings by failing to recognize this risk.

In sum, it is important to recognize that risks do not usually arise independently, but generally interact with one another, a problem that is even more critical in stressed market conditions. The resulting combined risk is practically always non-linear in that the total risk faced is worse than the sum of the risks of the separate components. Most risk models and systems do not directly account for risk interactions, which makes the consequences of the risk interaction even worse. Governance bodies, company management, and financial analysts should be keenly aware of the potential risk and damage of risks in combination, and be aware of the dangers of treating risks as separate and unrelated.

EXAMPLE 3

Financial and Non-Financial Sources of Risk

1. Which of the following is *not* a financial risk?

A. Credit risk
B. Market risk
C. Operational risk

Hide Solution

Solution

C is correct. Operational risk is the only risk listed that is considered non-financial, even though it may have financial consequences. Credit and market risks derive from the possibility of default and market movements, respectively, and along with liquidity risk, are considered financial risks.

2. Which of the following *best* describes an example of interactions among risks?

A. A stock in Russia declines at the same time as a stock in Japan declines.
B. Political events cause a decline in economic conditions and an increase in credit spreads.
C. A market decline makes a derivative counterparty less creditworthy while causing it to owe more money on that derivative contract.

Hide Solution

Solution

C is correct. Although most risks are likely to be interconnected in some way, in some cases the risks an organization is exposed to will *interact* in such a way that a loss (or gain) in one exposure will lead directly to a loss in a different exposure as well, such as with many counterparty contracts. Conditions in A and B are much more directly linked in that market participants fully expect what follows—for example, in B, an outbreak of war

for credit-risky securities. In C, in contrast, the reduction in creditworthiness following the market decline may be expected, but owing more money on an already existing contract as a result comes from the interaction of risks.

3. Which of the following *best* describes a financial risk?

A. The risk of an increase in interest rates.
B. The risk that regulations will make a transaction illegal.
C. The risk of an individual trading without limits or controls.

Hide Solution

Solution

A is correct because this risk arises from the financial markets.

4. Which of the following is *not* an example of model risk?

A. Assuming the tails of a returns distribution are thin when they are, in fact, fat.
B. Using standard deviation to measure risk when the returns distribution is asymmetric.
C. Using the one-year risk-free rate to discount the face value of a one-year government bond.

Hide Solution

Solution

C is correct. The risk-free rate is generally the appropriate rate to use in discounting government bonds. Although government bonds are generally default free, their returns are certainly risky. Assuming a returns distribution has thin tails when it does not and assuming symmetry in an asymmetric distribution are both forms of model risk.

5. Which of the following is the risk that arises when it becomes difficult to sell a security in a highly stressed market?

A. Liquidity risk
B. Systemic risk
C. Wrong-way risk

Hide Solution

Solution

A is correct. Securities vary highly in how liquid they are. Those with low liquidity are those for which either the number of agents willing to invest or the amount of capital these agents are willing to invest is limited. When markets are stressed, these limited number of investors or small amount of capital dry up, leading to the inability to sell the security at any price the seller feels is reasonable. Systemic risk is the risk of failure of the entire financial system and a much broader risk than liquidity risk. Wrong-way risk is the extent to which one's exposure to a counterparty is positively related to the counterparty's credit risk.

6. The risks that individuals face based on mortality create which of the following problems?

A. The risk of loss of income to their families.
B. Covariance risk associated with their human capital and their investment portfolios.
C. The interacting effects of solvency risk and the risk of being taken advantage of by an unscrupulous financial adviser.

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Solution

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by a termination of the income stream generated by the person. In contrast, longevity risk is the risk of outliving one's financial resources.

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